

Joyce Berdkan

Vienna, Virginia | (434) 333-9336 | joyceberdkan@hotmail.com |
<https://www.linkedin.com/in/joyce-berdkan/> | <https://jberdkan.github.io/My-Portfolio/> |
<https://github.com/jberdkan>

EDUCATION

Master of Science, Computer Engineering

August 2025 - May 2027

George Washington University, Washington, DC

Field of Focus: Electronics, Photonics, and MEMs

M.S. Thesis: Ring-oscillator Potts machine for max-3-cut in SkyWater 130 nm – coupled current-starved ring oscillators with third-harmonic injection locking; advised by Dr. Weidong Cao (see Research)

Relevant Coursework: Device Electronics, ASIC Design, High-Level VLSI Design

Bachelor of Science, Computer Engineering

August 2021 - May 2025

Liberty University, Lynchburg, VA

Minor: Mathematics

GPA: 3.67/4.0

Academic Honors: Magna Cum Laude

Clubs & Organizations: Society of Women Engineers, IEEE, National Arab American

Association of Engineers & Architects

Relevant Coursework: Computer Architecture, Real-Time Embedded Systems, Computer Networks, Algorithms and Data Structure

RESEARCH

Ring-Oscillator Potts Machine for Max-3-Cut – SkyWater 130 nm

May 2026 – Present

George Washington University, M.S. Thesis (co-authored publication in preparation)

- Studying a ring-oscillator Potts machine in the open-source SkyWater sky130 PDK that targets max-3-cut rather than the usual max-cut; all results so far come from transistor-level SPICE simulation, checked against an exact ILP solver.
- Coupled current-starved ring oscillators are pinned to three phase states by third-harmonic injection locking, and RC resistive couplers encode the edge weights.
- Characterized the injection lock range and how it scales with injection current, and the coupling's precision limits: roughly 8 bits of dynamic range, with a robustness cliff near 3% resistor mismatch.
- Identified a fragility on a frustrated 8-node instance: a $\pm 0.01\%$ shift in injection frequency flips 8 of 20 initial conditions between the optimum and a trapped two-coloring.
- Showed that noise confined to the injection ramp removes this sensitivity (19/20 optimal, versus 6–10/20 without) and that raising injection amplitude does not substitute for it; now testing whether this holds beyond a single instance.

EXPERIENCE

R&D Electrical Engineer Intern

May 2026 – July 2026

Solakair, Fremont, CA

- Independently led root-cause investigation into a persistent actuator suppression fault by instrumenting the flight-control source and cross-referencing data flash logs, systematically eliminating the sensor-interface layer to isolate the defect to internal control logic.

- Diagnosed a fixed-wing flight instability to a single lift-curve-slope coefficient error in the simulation's aerodynamic model; validated the correction across the full flight envelope (hover, transition, loiter, RTL, landing).
- Built a full nonlinear flight dynamics model for a novel two-tilt-rotor airframe, integrating CFD-derived aerodynamic data and motor thrust curves; resolved a coordinated-convention mismatch between supervisor and simulation-standard reference frames.
- Achieved PID/autotune convergence across all three control axes in simulation via a scheduler timing correction.
- Coordinated task assignments and progress review for three interns on a parallel simulation effort at supervisor's request.

Geographic Information System (GIS) Research
 Liberty University Facilities Information Management
 Lynchburg, VA

June 2024 – April 2025

- Collected, processed, and entered spatial data from various sources into GIS databases
- Managed and maintained up-to-date GIS databases, ensuring accuracy and reliability of spatial and non-spatial data
- Created detailed maps, charts, and visualizations to support geographic data analysis and project goals
- Performed spatial analysis using GIS software (ArcGIS) for urban planning, environmental studies, and transportation projects
- Designed and implemented GIS projects, contributing to data analysis and map production

Undergraduate Student Researcher
 Liberty University, Traffic Simulator Research
 Lynchburg, VA

September 2023 – May 2024

- Conduct research under a Liberty University faculty member, Dr. Songsu Son, on a traffic simulator in the US.
- Create and develop autonomous vehicle on the RealTime Car Simulator
- Create simulation on RealTime Technologies Traffic Simulator using the given programs.
- Program new blocks on C/C++ and JavaScript to further the research on the simulation.

Undergraduate Student Researcher
 Liberty University, UAV Technology Research
 Lynchburg, VA

June 2023 – May 2024

- Conduct research under a Liberty University faculty member, Dr. Songsu Son, on using UAVs in developing countries
- Data analyzing and researching various thermal camera technologies, photogrammetry, GPS systems, and power systems that can be used within a UAV

ADDITIONAL EXPERIENCE

Student Technical Support Assistant I
 Division of Information Technology (VSTC)
 George Washington University, Washington, DC

October 2025 – Present

- Provide basic technical and computer support to faculty, staff, and students through in-person and remote assistance.
- Respond to customer inquiries and troubleshoot issues in classrooms, conference rooms, and office environments to ensure technology is functional and available.
- Create, modify and resolve IT support tickets while coordinating with the IT Support Center (ITSC) to address escalated issues.

- Assist with setup, testing and maintenance of instructional and administrative technology across the School of Nursing and Education Centers.
- Deliver clear information on GW IT services and resources to the university community.
- Support special IT projects and departmental initiatives to improve service delivery and operational efficiency.

Laboratory Technician

August 2021 – May 2023

Liberty University, School of Engineering
Lynchburg, VA

- Acted as Team Lead of the laboratory's Computer Services department
- Troubleshoot and resolve hardware and software issues for end-users
- Installed and configured engineering applications and updates
- Maintain inventory of hardware and software assets and ordered new equipment as needed
- Helped in leading team meetings and assisted the Team Lead in the distribution of tasks

Controls and Propulsion

August 2022 – May 2023

Liberty University Space – Aero Competition Team (SAE)

- Design, build, and fly fixed-wing, remotely piloted aircraft for collegiate aeronautical competitions
- Design and manufacture all the controls and radio control systems on the unmanned aircraft
- Design the battery to be the same qualification as stated in the general aircraft requirements.

Electrical Engineer Motorsports

January 2022-May 2022

Liberty Motorsports Formula Society of Automotive Engineers (SAE)

- Compete to test race car's speed, durability, endurance, cost, and creativity
- Design and manufacture the electrical components for a Formula One-style race car

PROJECTS

8-bit SAR ADC in SkyWater 130 nm – Design, Layout & RTL-to-GDS Aug 2026 – Present

Independent Project – IEEE SSCS Code-a-Chip submission, ISSCC 2027

Open-source Jupyter notebooks (Apache-2.0), reproducible on free Google Colab; typical corner, mismatch disabled

- Designed an 8-bit charge-redistribution SAR ADC in the sky130A PDK – StrongARM comparator, binary-weighted MiM capacitor DAC (256×4.83 fF units in a dispersed common-centroid pattern), and one-hot SAR sequencer – simulated at transistor level (774 devices) in ngspice at 1 MS/s from a 1.8 V supply.
- Measured 43.5–43.9 dB SNDR and 49.0–49.7 dB SFDR on three coherently sampled tones at $0.93\text{--}0.99 \times$ Nyquist (7.11–7.17 ENOB, referred to full scale from a -1 dBFS input); across a 520-point ramp, 0.31 LSB rms / 0.73 LSB peak INL with all 238 codes present and zero inversions.
- Root-caused a persistent sub-7-bit ENOB to a sequencer timing collision that cleared the code register before the LSB was captured (the output never produced an odd code); replaced the hardcoded clear with timing derived from the bit count and, with a solver-tolerance fix, raised ENOB at Nyquist from 6.57 to 7.16 bits.
- Showed that ngspice's default tolerance (reltol = $1e-3$) mis-coded 7 of 28 test inputs by up to 2 LSB and that a 40-sample record biases ENOB above the 8-bit theoretical ceiling; validated the SNDR scorer on ideal and stuck-bit reference converters before trusting circuit results.
- Generated the MiM capacitor array layout in Python (gdstk) at the $3.0\text{ }\mu\text{m}$ pitch set by the sky130 capm spacing rules; the routed 4-bit common-centroid demonstrator is clean against the

full sky130A_mr.drc deck and LVS-matched in KLayout, with unit capacitance fitted from area and perimeter terms.

- Rewrote the SAR sequencer as synthesizable Verilog verified in Icarus Verilog; diffing the RTL against the transistor netlist caught an MSB sampling-phase defect that the ideal-DAC testbench cannot detect. Implemented it through OpenROAD on sky130hd: 78 cells, +8.61 ns worst slack, 31.2 μ W, 0 violations on the full sign-off DRC deck.
- Scaled the design to a 16-channel controller (432 flip-flops, per-channel self-checking testbench) and used it to replicate the OpenLayout agentic physical-design framework (ICCAD '26), documenting five reproducibility defects in its open-source release.

Open-Source SIMD GPU – Full ASIC Implementation & TinyTapeout

July 2026

Independent Project – github.com/jberdikan/tiny-gpu

Original GPU architecture, ISA, and RTL credited to open source author adam-maj; this work is an independent physical-design and DFT extension

- Converted the SIMD GPU RTL from SystemVerilog to portable Verilog 2005, rewriting the memory controller's variable-index writes into a Yosys-safe form to run the design through both commercial and open-source ASIC toolchains.
- Ran a full RTL-to-GDSII flow in Cadence – Design Compiler synthesis, TetraMax scan insertion and ATPG (99.91% stuck-at, 98.07% transition-delay coverage), and Innovus place-and-route on OSU 0.5 μ m – zero DRC and LVS violations at signoff.
- Built a TinyTapeout adapter fitting the design to an 8-in/8-out pin interface via on-chip program/data memory and a byte-serial host loader, plus a shrunk 1-core/2-thread configuration; verified the matadd kernel end-to-end through the pin protocol.
- Hardened the adapter on sky130 via LibraLane to DRC/LVS/antenna-clean tapeout deliverables (177,166 μ m², 6x2 tiles), producing a fabricable GDSII.

Tiny MIPS CPU – Full ASIC Design Flow

Spring 2026

George Washington University, ASIC Design Course

Washington, DC

- Designed and implemented a multi-cycle Tiny MIPS CPU in Verilog, executing a ten-instruction subset (add, sub, and, or, slt, addi, beq, j, lb, sb) through a 13-state finite-state-machine controller sequencing a shared 8-bit datapath across fetch, decode, execute, memory and write-back stages.
- Completed the full back-end ASIC flow – logic synthesis, design-for-test scan-chain insertion, ATPG, place-and-routed, and pad-frame integration – using industry-standard Synopsys and Cadence EDA tools.
- Achieved 99.49% stuck-at fault coverage with 122 ATPG patterns and 98.09% transition-delay fault coverage with 318 patterns using Synopsys TetraMAX.
- Verified functional correctness with a self-checking testbench reporting full instruction-class coverage, confirmed by post-route gate-level static timing closure.
- Closed timing at 25 MHz with zero violating paths across all 439 timed endpoints; placed-and-routed core occupies 0.825 mm² in the OSU 0.5 μ m technology at an estimated 49 mW.
- Integrated the core into a Frame2h_68 pad-frame from the UofU_Pads library using Cadence Chip Assembly Router, auto-routing all 31 signal pads without conflicts.

64-QAM Digital Modulator with SPI Interface and CDC FIFO

Spring 2026

George Washington University, High-Level VLSI Design Course

Washington, DC

- Designed and implemented. 64-QAM digital modulator in Verilog featuring a 64-QAM symbol mapper, an SPI configuration interface, and a synchronous-FIFO clock-domain-crossing (CDC) bridge with three-domain reset sequencing.

- Synthesized the design with the IBM CMOS8HP standard-cell library at the slow-slow, -125°C corner, meeting timing at approximately 130 MHZ maximum frequency with a total cell area near 812,000 μm^2 .
- Diagnosed and resolved X-propagation in back-annotated gate-level simulation by converting negative-edge-triggered flip-flops to positive-edge logic using active-high intermediate signals.
- Performed SDF-annotated, timing-accurate gate-level simulation in Cadence Xcelium, debugging SPI task timing, testbench capture windows, and \$sdf_annotate scoping.
- Validated end-to-end modulator behavior against a self-checking testbench across both the synthesized and back-annotated netlists.

Resistive Training Device

Fall 2024 – Spring 2025

Liberty University, Engineering Senior Design Project
Lynchburg, VA

- Collaborating with a multidisciplinary engineering team on a capstone project for Club Sports Swimming Team at Liberty University on implementing and designing a new training device for swimmers.
- Designed and implemented a real-time embedded system using FreeRTOS on the STM32F412 board.
- Programmed and calibrated a variable Pulse Width Modulation control system driven by a potentiometer to modulate motor resistance based on swimmer feedback.
- Integrated an HX711 load cell amplifier and a 3-pulley tension sensor for force measurement, calculating real-time output in Newtons.
- Configured a hall effect sensor A3144E to measure drum rotations, enabling accurate distance and velocity tracking in meters and m/s.
- Developed a Finite State Machine (FSM) to manage system states, including active training, idle and reset modes.
- Implemented task synchronization and inter-process communication using FreeRTOS features like semaphores and soft timers.

FPGA PID Temperature Control System

Spring 2025

Liberty University, Advanced Embedded Systems Course
Lynchburg, VA

- Designed and implemented a hardware -accelerated PID (Proportional-Integral-Derivative) controller using Verilog for real-time temperature regulation of an FPGA chip on the DE1-SoC board.
- Built a custom PID IP core wrapped in Avalon bus interface and integrated it into a Qsys-extended SoC system (Platform Designer) containing ADC and PWM IP cores.
- Developed a complete embedded control system to regulate the temperature using a DC fan and temperature sensor, with feedback and actuation controlled by the PID logic.
- Programmed the software side on HPS in C to interface with the hardware accelerator and manage control feedback loops.
- Utilized Quartus and Platform Designer to simulate, test, and verify the complete PID system in both software and hardware.
- Produces a full technical report evaluating controller performance and demonstrating understanding of complex engineering problem-solving through ABET outcomes.

Single-cycle ARM Processor

Fall 2024

Liberty University, Computer Architecture Course
Lynchburg, VA

- Designing and implementing a single-cycle ARM processor using Verilog, capable of executing key instructions such as ADD, SUB, AND, ORR, LDR, STR, and B.
- Collaborating with a partner to build a fully functional processor, simulating and testing performance using Quartus and ModelSim.
- Loading and testing programs on the processor, verifying instruction execution and memory write operations through simulation waveforms and performance metrics.
- Applying key principles of computer architecture, including instruction decoding, ALU operations, and memory interfacing, to optimize processor functionality.

Microcontroller Car

Spring 2024

Liberty University, Real-Time Embedded Systems Course

Lynchburg, VA

- Designed and implemented a real-time embedded system using FreeRTOS on the STM32F429-Discovery board for a toy car.
- Developed and integrated four RTOS tasks to control the car's functions, including ultrasonic sensors and motor operations.
- Implemented soft timers for motor control and utilized mutexes and semaphores for task synchronization and resource management.
- Conducted Rate Monotonic (RM) schedulability analysis to evaluate task priority and system performance.
- Configured ultrasonic sensors for accurate distance measurement, triggering corresponding motor actions and LCD color changes based on proximity.
- Managed system states using a Finite State Machine (FSM) to control car movement and user-defined distance thresholds.

Common Emitter (CE) Printed Circuit Board (PCB)

Fall 2023

Liberty University, Electronics Course

Lynchburg, VA

- Collaborated with a team to design, manufacture, and test a dual-stage common-emitter (CE) BJT printed circuit board (PCB).
- Conducted research and analysis of a dual-stage BJT schematic, calculating key values such as gain, resistor, and capacitor values to achieve the desired total gain.
- Utilized MultiSim to simulate circuit performance and calculated theoretical values, refining design parameters to match project requirements.
- Designed and laid out a 2-layer PCB in KiCad, assigning footprints to electrical components and generating Gerber and drill files for manufacturing.
- Sourced components from open-source libraries, modifying the design to accommodate component availability, including resistor adjustments to meet specified values.
- Successfully soldered electrical components onto the PCB.
- Tested the assembled PCB, identifying distortion issues in the waveform due to noise, component tolerances, and design limitations.

Visible Light Communication (VLC) System

Spring 2024

Liberty University, Communications Course

Lynchburg, VA

- Designed and implemented a communication (VLC) system that transmits/receives audio signals via visible light as a means of a carrier.

- Input audio from a device (e.g. phone/computer) passed through a transistor circuit to drive the LED.
- Optimized the receiver with a phototransistor and LM386 amplifier for better signal reception.
- Addressed noise and interference by modifying circuit components with capacitors and resistors.

GRANTS, AWARDS, & SCHOLARSHIPS

Graduate Impact Award Fall 2025
George Washington University, School of Engineering and Applied Sciences

ONE Fellowship 2022 - 2025
Liberty University, Office of Opportunity and Enrichment

PROFESSIONAL AFFILIATIONS & MEMBERSHIPS

Society of Women Engineers May 2022 – Present
Treasurer May 2023-December 2024

- Maintained records of all section expenditures and income.
- Attended all required Student Government Association (SGA) meetings with the president, submitted chapter budget and kept track of the chapter bank account.
- Submitted annual financial report to national SWE.

National Arab American Association of Engineers & Architects May 2024 – Present

IEEE May 2024 – Present

COMMUNITY OUTREACH

SWE Women's Shelter Volunteer Spring 2023
YWCA
Lynchburg, VA

- Helps in cleaning, organizing inventory, and storing donations at the YWCA's women's shelter.
- Collaborated with a team in organizing a campus-wide clothing drive for the YWCA's donation program.

REFERENCES

Dr. Weidong Cao
George Washington University
Electrical Engineering
Faculty Professor
(202)-994-0810
weidong.cao@email.gwu.edu

Dr. Songsu Son
Liberty University
Civil Engineering Faculty
Associate Professor
(434) 592-2079
sson7@liberty.edu

Zhiqiang Yi
George Washington University
PhD student
zhiqiang.yi@gwu.edu